Michael Chen

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https://michaelkchen.github.io/

Education

Purdue University — **Expected May 2027**
B.S. in Computer Science; Minor in Art & Design Studio — *West Lafayette, IN*

- **Cumulative GPA:** 3.94 | **Dean's List & Semester Honors:** 6/6 Semesters
- **Relevant Coursework:** Analysis of Algorithms, Artificial Intelligence, Numerical Methods, Fundamentals of Computer Graphics, Data Structures And Algorithms, Computer Architecture, Systems Programming, Programming In C, Discrete Math, Problem Solving and Object Oriented Programming

Experience

Itron Inc. — **May 2026 – Aug. 2026**
Full Stack Developer Intern — *San Jose, CA*

SeeFood - Stanford Mike Snyder Lab Research (Abstract) — **May 2025 – Aug. 2025**
Software Engineer Intern — *Palo Alto, CA*

- Cooperated with multiple teams, including several professional research scientists, mobile app developers, and biology students with the overall goal of developing reliable, easy-to-use, and accurate technology for logging nutritional data for a wide range of meals.
- Co-authored a poster for research comparing GPT-4o-mini and SnapCalorie on their effectiveness of identifying ingredients and quantifying macronutrients of **35 total smartphone images of meals**, concluding that even though GPT-4o-mini is more accurate, **neither model yielded high accuracy at all.**
- Directed an independent focus, experimenting with image capturing and transferring utilizing Meta Ray-Ban Smartglasses and writing a Python script to feed **35 images** into OpenAI's Python library, optimizing recognition of ingredients for consistent results.

SLC - Stanford Mike Snyder Lab Research — **May 2024 – Aug. 2024**
Software Engineer Intern — *Palo Alto, CA*

- Collaborated with a team, including an assistant professor and graduate student, to analyze the effects of a Chinese herbal medicine on long COVID, based on a study involving wearable watch data from **19 patients over 8 weeks**.
- Surveyed **20GB** of time-series data on vital signs (heart rate, sleep, temperature), performing feature extraction in Python using tsfresh and catch22 to support machine learning applications.
- Created PCA plots with scikit-learn and Matplotlib, filtering data by time and patient to visualize trends over time, ultimately **finding no significant effects** of the medicine on patients' basic vitals.

Projects

Personal Website (GitHub) — **Apr. 2026 – Present**

- Constructed a website with the Astro framework utilizing HTML5, JavaScript, TypeScript, CSS, Markdown, and MDX, which is deployed on GitHub Pages with GitHub Actions.
- Designed a elegant, dynamic, and interactive thematic with graphics powered by the shaders.com library that provides accommodation for lower end machines.

Gaming Bot - Discord Chatbot (GitHub) — **Aug. 2018 – Present**

- Independently developed and managed a Discord chatbot, deploying it to a server with **150 members**, enabling users to play games, listen to music, moderate servers, and use other tools such as a poll creator.
- Ensured long term functionality and user satisfaction by monitoring frameworks and libraries, updating code regularly, modernizing UI/UX, and discarding irrelevant features to keep up with Discord's fast-pace evolution.
- Published to a GitHub repository, coded in JavaScript within a Node.js environment, interacts with users via an intuitive Markdown UI, hosted externally on Oracle Cloud, and utilizing a database with JSON files, previously hosted on Heroku with a PostgreSQL database.

Additional

Programming Languages: Python, C, C++, Java, JavaScript, TypeScript, LaTeX, Markdown, Matlab, HTML5, CSS
Frameworks/Libraries: tsfresh, scikit-learn, Matplotlib, catch22, Node.js, Discord.js, Pandas, NumPy, OpenGL, Astro
Infrastructure: Linux, Oracle Cloud, PostgreSQL, Heroku, Git, GitHub
Awards: USA Coding Olympiad Gold Status, Homestead High School Valedictorian